OS-JACKFRUIT

Multi-Container Runtime

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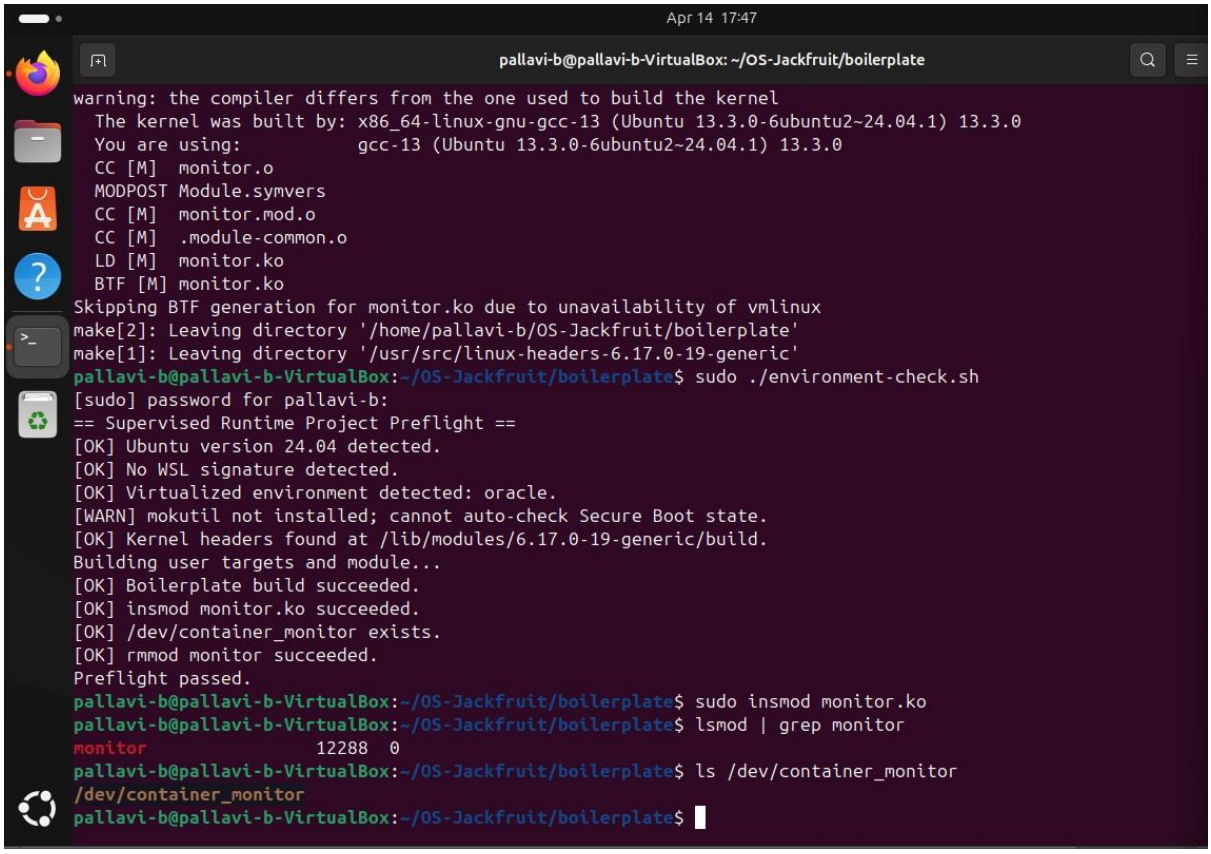
SEC: G

Screenshots:



TASK 1:

```
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate/rootfs$ ls
alpine-minirootfs-3.23.3-x86_64.tar.gz  dev  home  media  opt  root  sbin  sys  usr
bin                                     etc  lib   mnt    proc run   srv   tmp  var
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate/rootfs$ cd ..
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo cp memory_hog cpu_hog io_pulse rootfs/
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ ls rootfs/
alpine-minirootfs-3.23.3-x86_64.tar.gz  cpu_hog  etc   io_pulse  media       mnt  proc  run   srv  tmp  var
bin                                     dev      home  lib       memory_hog  opt  root  sbin  sys  usr
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine supervisor rootfs
[supervisor] ready (rootfs=rootfs socket=/tmp/mini_runtime.sock)
[supervisor] started 'c1' pid=9289
[supervisor] started 'c2' pid=9340
```

```
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine start c1 rootfs "cpu_hog 20"
[sudo] password for pallavi-b:
ok
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine start c2 rootfs "cpu_hog 20"
ok
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
c2               pid=9340   state=exited   soft= 40MiB hard= 64MiB code=127 sig=0   start=18:00:26
c1               pid=9289   state=exited   soft= 40MiB hard= 64MiB code=127 sig=0   start=17:59:40
```

```
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine start c4 rootfs "/cpu_hog 100"
ok
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
c4               pid=15111  state=running  soft= 40MiB hard= 64MiB code=0   sig=0   start=18:17:11
c3               pid=11641  state=exited   soft= 40MiB hard= 64MiB code=0   sig=0   start=18:15:28
c2               pid=10204  state=exited   soft= 40MiB hard= 64MiB code=0   sig=0   start=18:08:35
c1               pid=10180  state=exited   soft= 40MiB hard= 64MiB code=0   sig=0   start=18:08:25
c2               pid=9340   state=exited   soft= 40MiB hard= 64MiB code=127 sig=0   start=18:00:26
c1
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$
```

TASK 2:

```
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
c2               pid=10204  state=exited   soft= 40MiB hard= 64MiB code=0   sig=0   start
c1               pid=10180  state=exited   soft= 40MiB hard= 64MiB code=0   sig=0   start
c2               pid=9340   state=exited   soft= 40MiB hard= 64MiB code=127 sig=0   start
c1               pid=9289   state=exited   soft= 40MiB hard= 64MiB code=127 sig=0   start

pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ ls   rootfs/
alpine-minirootfs-3.23.3-x86_64.tar.gz  cpu_hog  etc   io_pulse  media       mnt  proc  r
bin                                     dev      home  lib       memory_hog  opt  root  s
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine logs c1
/bin/sh: cpu_hog: not found
cpu_hog alive elapsed=1 accumulator=8488015175725468995
cpu_hog alive elapsed=2 accumulator=6033449358966712512
cpu_hog alive elapsed=3 accumulator=4339841198134536119
cpu_hog alive elapsed=4 accumulator=7292795584196533636
cpu_hog alive elapsed=5 accumulator=8615908445189209954
cpu_hog alive elapsed=6 accumulator=2397641973987662503
cpu_hog alive elapsed=7 accumulator=17675932111116908633
cpu_hog alive elapsed=8 accumulator=12075446813871187839
cpu_hog alive elapsed=9 accumulator=11390401373559937977
cpu_hog alive elapsed=10 accumulator=6427755514765479242
cpu_hog alive elapsed=11 accumulator=5547866628412097498
cpu_hog alive elapsed=12 accumulator=13225349365678508617
cpu_hog alive elapsed=13 accumulator=14462013591419234240
cpu_hog done duration=20 accumulator=6801000933385675702
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine stop c1
container not found or not running
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$
```

TASK 3:

```
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine run c3 rootfs "/cpu_hog 5"
[run] waiting for container to finish…
state=exited code=0 signal=0
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ cat logs/c3.log
cpu_hog alive elapsed=1 accumulator=6714212733155315941
cpu_hog alive elapsed=2 accumulator=6618240562122467652
cpu_hog alive elapsed=3 accumulator=1623912300560331359 6
cpu_hog alive elapsed=4 accumulator=9514322129083419 71
cpu_hog alive elapsed=5 accumulator=1810959324513087142 6
cpu_hog alive elapsed=6 accumulator=6861087039228806 81
cpu_hog alive elapsed=7 accumulator=3401053677491740527
cpu_hog alive elapsed=8 accumulator=1223303865565896653 7
cpu_hog alive elapsed=9 accumulator=2041230254743977408
cpu_hog alive elapsed=13 accumulator=7170637677605683577
cpu_hog alive elapsed=14 accumulator=16393574937059400447
cpu_hog alive elapsed=15 accumulator=11827388193847270057
cpu_hog alive elapsed=16 accumulator=14377626064443733320
cpu_hog alive elapsed=17 accumulator=13522037886052027350
cpu_hog alive elapsed=18 accumulator=7607683809324154061
cpu_hog alive elapsed=19 accumulator=12651013183571407044
cpu_hog alive elapsed=20 accumulator=2518993587427591787
cpu_hog alive elapsed=21 accumulator=4434451414469635023
cpu_hog alive elapsed=22 accumulator=13965148304984013561
```

```
Apr 14 18:21
pallavi-b@pallavi-b-VirtualBox: ~/OS-Jackfruit/boilerplate
pallavi-b@pallavi-b-VirtualBox: ~/OS-Jackfruit/boilerplate   pallavi-b@pallavi-b-VirtualBox: ~/OS-Jackfruit/boilerplate
cpu_hog alive elapsed=25 accumulator=6534811994822408123
cpu_hog alive elapsed=26 accumulator=6760259344068015841
cpu_hog alive elapsed=30 accumulator=1147220970640167515
cpu_hog alive elapsed=31 accumulator=5364839943822199549
cpu_hog alive elapsed=32 accumulator=7264092236270666038
cpu_hog alive elapsed=33 accumulator=4499155384538955452
cpu_hog alive elapsed=34 accumulator=14639179011983481384
cpu_hog alive elapsed=35 accumulator=12865104147413858319
cpu_hog alive elapsed=36 accumulator=12712876961592311047
cpu_hog alive elapsed=37 accumulator=1128544247641671581
cpu_hog alive elapsed=38 accumulator=386858397204211678
cpu_hog alive elapsed=39 accumulator=6015178973589781073
cpu_hog alive elapsed=54 accumulator=15006476381075981262
cpu_hog alive elapsed=55 accumulator=17918464280488886155
cpu_hog alive elapsed=56 accumulator=632456040251411127
cpu_hog alive elapsed=57 accumulator=7384659404389094684
cpu_hog alive elapsed=58 accumulator=1096823671401906327
cpu_hog alive elapsed=59 accumulator=9557075351983169063
cpu_hog alive elapsed=60 accumulator=411175023313787733
cpu_hog alive elapsed=61 accumulator=7619184934965498454
cpu_hog alive elapsed=62 accumulator=1502981503678653814
cpu_hog alive elapsed=63 accumulator=1768525071273811290
cpu_hog alive elapsed=64 accumulator=5842404482700947076
cpu_hog alive elapsed=65 accumulator=4295718307275054603
cpu_hog alive elapsed=79 accumulator=13487021162472607042
cpu_hog alive elapsed=80 accumulator=10544385020702347777
cpu_hog alive elapsed=81 accumulator=1704060486756370775
cpu_hog alive elapsed=82 accumulator=13684449482333565960
cpu_hog alive elapsed=83 accumulator=3608759285278802025
cpu_hog alive elapsed=84 accumulator=11782575117011700673
cpu_hog alive elapsed=85 accumulator=14185679886525056786
```

TASK 4:

```
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine run m1 rootfs "/memory_hog 8 500"
[run] waiting for container to finish…
state=killed code=0 signal=9
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ dmesg | tail
dmesg: read kernel buffer failed: Operation not permitted
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo dmesg | tail
[ 4258.085869] [container_monitor] pid=16902 (container=c3) gone, removing entry
[ 4292.143385] workqueue: vmstat_update hogged CPU for >10000us 7 times, consider switching to WQ_UNBOUND
[ 4380.408484] [container_monitor] Registered container=m1 pid=17838 soft=41943040 hard=67108864
[ 4382.867403] [container_monitor] SOFT LIMIT container=m1 pid=17838 rss=42409984 limit=41943040
[ 4384.915134] [container_monitor] HARD LIMIT container=m1 pid=17838 rss=75964416 limit=67108864 - SIGKILL sent
[ 4393.101766] workqueue: delayed_vfree_work hogged CPU for >10000us 7 times, consider switching to WQ_UNBOUND
[ 4434.611571] audit: type=1400 audit(1776171183.890:414): apparmor="DENIED" operation="open" class="file" profile="snap
.firefox.firefox" name="/proc/pressure/memory" pid=4141 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=100
0 ouid=0
[ 4444.967648] audit: type=1400 audit(1776171194.251:415): apparmor="DENIED" operation="open" class="file" profile="snap
.firefox.firefox" name="/proc/pressure/memory" pid=4141 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=100
0 ouid=0
[ 4482.619376] audit: type=1400 audit(1776171231.904:416): apparmor="DENIED" operation="open" class="file" profile="snap
.firefox.firefox" name="/proc/pressure/memory" pid=4141 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=100
0 ouid=0
[ 4487.622521] audit: type=1400 audit(1776171236.907:417): apparmor="DENIED" operation="open" class="file" profile="snap
.firefox.firefox" name="/proc/pressure/memory" pid=4141 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=100
0 ouid=0
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$
```

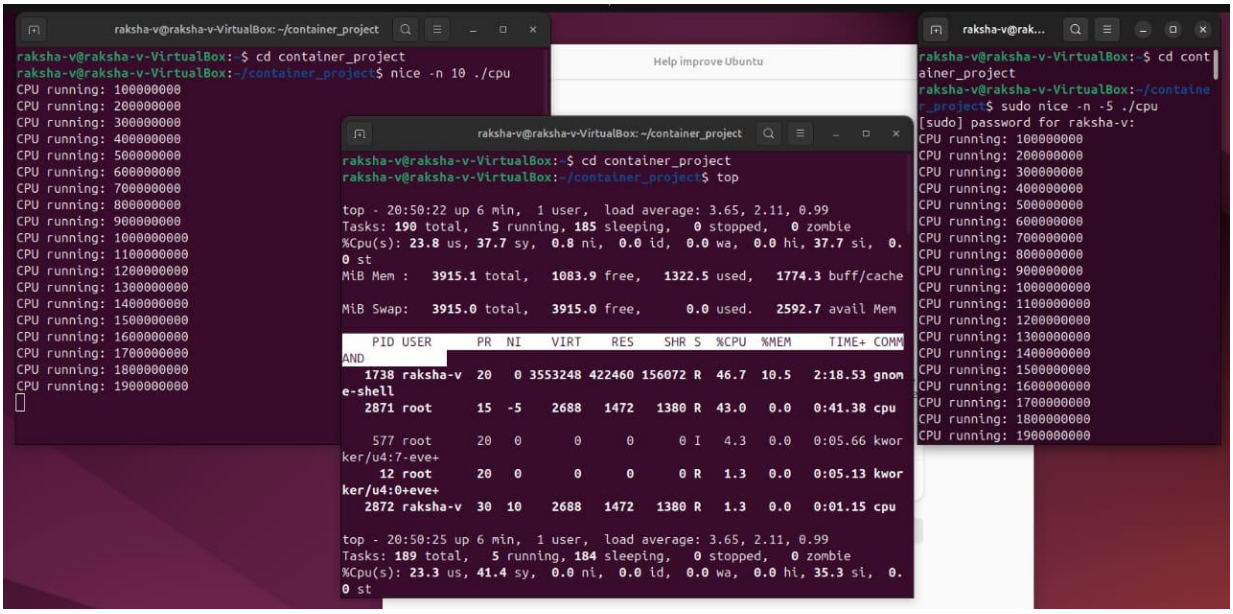
TASK 5:

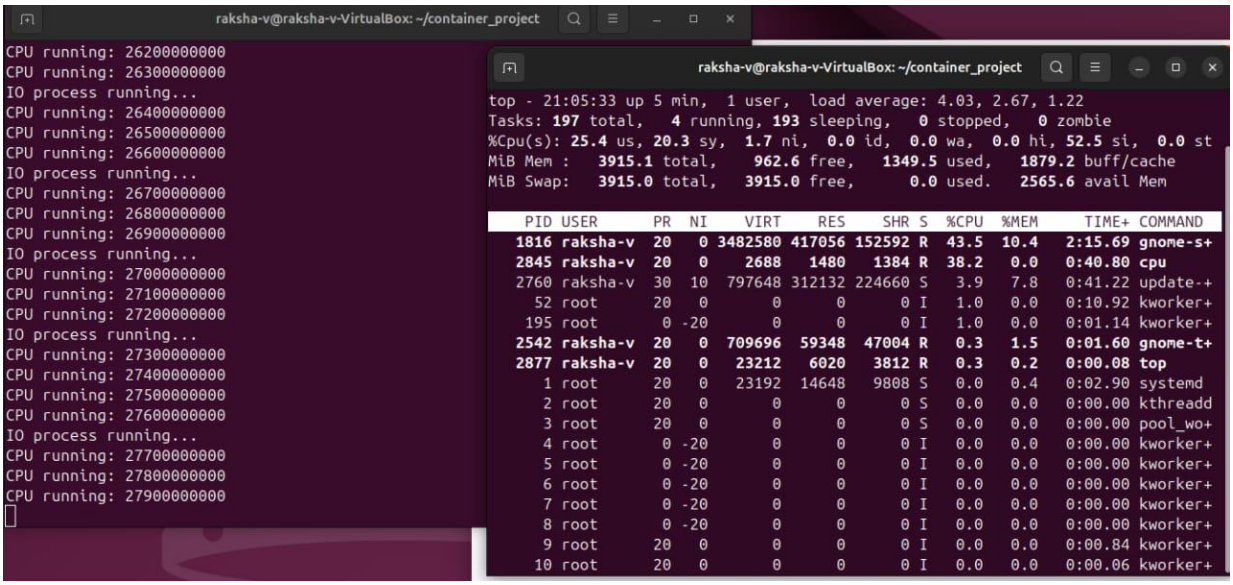
```
CPU running: 2500000000
CPU running: 2600000000
IO process running...
CPU running: 2700000000
IO process running...
CPU running: 2800000000
CPU running: 2900000000
IO process running...
CPU running: 3000000000
CPU running: 3100000000
CPU running: 3200000000
IO process running...
CPU running: 3300000000
CPU running: 3400000000
CPU running: 3500000000
IO process running...
CPU running: 3600000000
CPU running: 3700000000
CPU running: 3800000000
IO process running...
CPU running: 3900000000
CPU running: 4000000000
IO process running...
```

```
1234: command not found

[5]+  Stopped                 sudo nice -n -5 ./cpu
raksha-v@raksha-v-VirtualBox:~/container_project$ sudo nice -n -5 ./cpu
[sudo] password for raksha-v:
CPU running: 100000000
CPU running: 200000000
CPU running: 300000000
CPU running: 400000000
CPU running: 500000000
CPU running: 600000000
CPU running: 700000000
CPU running: 800000000
CPU running: 900000000
CPU running: 1000000000
CPU running: 1100000000
CPU running: 1200000000
CPU running: 1300000000
CPU running: 1400000000
CPU running: 1500000000
```

```
raksha-v@raksha-v-VirtualBox:~$ cd ~/container_project
raksha-v@raksha-v-VirtualBox:~/container_project$ nice -n 10 ./cpu
CPU running: 100000000
CPU running: 200000000
CPU running: 300000000
CPU running: 400000000
CPU running: 500000000
CPU running: 600000000
```


raksha-v@raksha-v-VirtualBox: ~/container_project
raksha-v@raksha-v-VirtualBox:~$ cd container_project
raksha-v@raksha-v-VirtualBox:~/container_project$ nice -n 10 ./cpu
CPU running: 100000000
CPU running: 200000000
CPU running: 300000000
CPU running: 400000000
CPU running: 500000000
CPU running: 600000000
CPU running: 700000000
CPU running: 800000000
CPU running: 900000000
CPU running: 1000000000
CPU running: 1100000000
CPU running: 1200000000
CPU running: 1300000000
CPU running: 1400000000
CPU running: 1500000000
CPU running: 1600000000
CPU running: 1700000000
CPU running: 1800000000
CPU running: 1900000000
Help improve Ubuntu
raksha-v@raksha-v-VirtualBox: ~/container_project
raksha-v@raksha-v-VirtualBox:~$ cd container_project
raksha-v@raksha-v-VirtualBox:~/container_project$ top
top - 20:50:22 up 6 min, 1 user, load average: 3.65, 2.11, 0.99
Tasks: 190 total, 5 running, 185 sleeping, 0 stopped, 0 zombie
%Cpu(s): 23.8 us, 37.7 sy, 0.8 ni, 0.0 id, 0.0 wa, 0.0 hi, 37.7 si, 0.0 st
MiB Mem : 3915.1 total, 1083.9 free, 1322.5 used, 1774.3 buff/cache
MiB Swap: 3915.0 total, 3915.0 free, 0.0 used. 2592.7 avail Mem
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
1738 raksha-v 20 0 3553248 422460 156072 R 46.7 10.5 2:18.53 gnome-shell
2871 root 15 -5 2688 1472 1380 R 43.0 0.0 0:41.38 cpu
577 root 20 0 0 0 0 I 4.3 0.0 0:05.66 kworker/u4:7-eve+
12 root 20 0 0 0 0 R 1.3 0.0 0:05.13 kworker/u4:0+eve+
2872 raksha-v 30 10 2688 1472 1380 R 1.3 0.0 0:01.15 cpu
top - 20:50:25 up 6 min, 1 user, load average: 3.65, 2.11, 0.99
Tasks: 189 total, 5 running, 184 sleeping, 0 stopped, 0 zombie
%Cpu(s): 23.3 us, 41.4 sy, 0.0 ni, 0.0 id, 0.0 wa, 0.0 hi, 35.3 si, 0.0 st
raksha-v@rak...
raksha-v@raksha-v-VirtualBox:~$ cd container_project
raksha-v@raksha-v-VirtualBox:~/container_project$ sudo nice -n -5 ./cpu
[sudo] password for raksha-v:
CPU running: 100000000
CPU running: 200000000
CPU running: 300000000
CPU running: 400000000
CPU running: 500000000
CPU running: 600000000
CPU running: 700000000
CPU running: 800000000
CPU running: 900000000
CPU running: 1000000000
CPU running: 1100000000
CPU running: 1200000000
CPU running: 1300000000
CPU running: 1400000000
CPU running: 1500000000
CPU running: 1600000000
CPU running: 1700000000
CPU running: 1800000000
CPU running: 1900000000


raksha-v@raksha-v-VirtualBox: ~/container_project
CPU running: 26200000000
CPU running: 26300000000
IO process running...
CPU running: 26400000000
CPU running: 26500000000
CPU running: 26600000000
IO process running...
CPU running: 26700000000
CPU running: 26800000000
CPU running: 26900000000
IO process running...
CPU running: 27000000000
CPU running: 27100000000
CPU running: 27200000000
IO process running...
CPU running: 27300000000
CPU running: 27400000000
CPU running: 27500000000
CPU running: 27600000000
IO process running...
CPU running: 27700000000
CPU running: 27800000000
CPU running: 27900000000
top - 21:05:33 up 5 min, 1 user, load average: 4.03, 2.67, 1.22
Tasks: 197 total, 4 running, 193 sleeping, 0 stopped, 0 zombie
%Cpu(s): 25.4 us, 20.3 sy, 1.7 ni, 0.0 id, 0.0 wa, 0.0 hi, 52.5 si, 0.0 st
MiB Mem : 3915.1 total, 962.6 free, 1349.5 used, 1879.2 buff/cache
MiB Swap: 3915.0 total, 3915.0 free, 0.0 used. 2565.6 avail Mem
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
1816 raksha-v 20 0 3482580 417056 152592 R 43.5 10.4 2:15.69 gnome-s+
2845 raksha-v 20 0 2688 1480 1384 R 38.2 0.0 0:40.80 cpu
2760 raksha-v 30 10 797648 312132 224660 S 3.9 7.8 0:41.22 update-+
52 root 20 0 0 0 0 I 1.0 0.0 0:10.92 kworker+
195 root 0 -20 0 0 0 I 1.0 0.0 0:01.14 kworker+
2542 raksha-v 20 0 709696 59348 47004 R 0.3 1.5 0:01.60 gnome-t+
2877 raksha-v 20 0 23212 6020 3812 R 0.3 0.2 0:00.08 top
1 root 20 0 23192 14648 9808 S 0.0 0.4 0:02.90 systemd
2 root 20 0 0 0 0 S 0.0 0.0 0:00.00 kthreadd
3 root 20 0 0 0 0 S 0.0 0.0 0:00.00 pool_wo+
4 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker+
5 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker+
6 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker+
7 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker+
8 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker+
9 root 20 0 0 0 0 I 0.0 0.0 0:00.84 kworker+
10 root 20 0 0 0 0 I 0.0 0.0 0:00.06 kworker+

TASK 6:

```
./var/log/
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate/rootfs$ ls
alpine-minirootfs-3.23.3-x86_64.tar.gz  dev  home  media  opt   root  sbin  sys  usr
bin                                     etc  lib   mnt    proc  run   srv   tmp  var
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate/rootfs$ cd ..
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo cp memory_hog cpu_hog io_pulse rootfs/
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ ls rootfs/
alpine-minirootfs-3.23.3-x86_64.tar.gz  cpu_hog  etc   io_pulse  media       mnt  proc  run   srv  tmp  var
bin                                     dev      home  lib       memory_hog  opt  root  sbin  sys  usr
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine supervisor rootfs
[supervisor] ready (rootfs=rootfs socket=/tmp/mini_runtime.sock)
[supervisor] started 'c1' pid=9289
[supervisor] started 'c2' pid=9340
[supervisor] started 'c1' pid=10180
[supervisor] started 'c2' pid=10204
[supervisor] started 'c3' pid=11641
Container 'c3' already running
Container 'c3' already running
Container 'c3' already running
[supervisor] started 'c4' pid=15111
[supervisor] started 'c3' pid=16902
[supervisor] started 'm1' pid=17838
[supervisor] started 'c4' pid=18172
[supervisor] started 'c5' pid=18216
[supervisor] started 'c8' pid=18524
[supervisor] started 'c9' pid=18534
[supervisor] started 'c10' pid=18547
```